3.15 Timers and watchdogs

The device includes an advanced-control timer, four general-purpose timers, two low-power timers, two watchdog timers and a SysTick timer. *Table 7* compares features of the advanced-control, general-purpose and basic timers.

Table 7. Timer feature comparison

Timer	Timer type	Counter resolution	Counter type	Maximum operating frequency	Prescaler factor	DMA request generation	Capture/ compare channels	Comple-mentary outputs
TIM1	Advanced-control	16-bit	Up, down, up/down	48 MHz	Integer from 1 to 2^{16}	Yes	4 +2 internal	3
TIM3	General-purpose	16-bit	Up, down, up/down	48 MHz	Integer from 1 to 2^{16}	Yes	4	-
TIM14	General-purpose	16-bit	Up	48 MHz	Integer from 1 to 2^{16}	No	1	-
TIM16 TIM17	General-purpose	16-bit	Up	48 MHz	Integer from 1 to 2^{16}	Yes	1	1

3.15.1 Advanced-control timer (TIM1)

The advanced-control timer can be seen as a three-phase PWM unit multiplexed on 6 channels. It has complementary PWM outputs with programmable inserted dead-times. It can also be seen as a complete general-purpose timer. The four independent channels can be used for:

- input capture
- output compare
- PWM output (edge or center-aligned modes) with full modulation capability (0-100%)
- one-pulse mode output

On top of these, there are two internal channels that can be used.

In debug mode, the advanced-control timer counter can be frozen and the PWM outputs disabled, so as to turn off any power switches driven by these outputs.

Many features are shared with those of the general-purpose TIMx timers (described in *Section 3.15.2*) using the same architecture, so the advanced-control timers can work together with the TIMx timers via the Timer Link feature for synchronization or event chaining.

3.15.2 General-purpose timers (TIM3, 14, 16, 17)

There are four synchronizable general-purpose timers embedded in the device (refer to *Table 7* for comparison). Each general-purpose timer can be used to generate PWM outputs or act as a simple timebase.

- TIM3

  This is a full-featured general-purpose timer with 16-bit auto-reload up/downcounter and 16-bit prescaler.

  It has four independent channels for input capture/output compare, PWM or one-pulse mode output. It can operate in combination with other general-purpose timers via the